Cell structure

CELLS ARE THE BUILDING BLOCKS OF LIFE.

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The cell is the basic unit of living things, with many millions working together to form an individual organism. Each cell is an enclosed sac containing everything it needs to survive and do its job.

Animal cell

The average animal cell grows to about 10 μm across (a 100th of a millimeter) although single cells inside eggs, bones, or muscles can reach several centimeters across. Animal bodies contain a large number of cell types, each specialized to do different jobs. Some kinds of single-celled protists, such as amoebas and protozoans, have a cell body very similar in structure to the cells of animals.

Centrosome
This produces long and thin strands used for hauling objects around the cell.

Cytoplasm
A watery filling of the cell with minerals dissolved in it.

Mitochondrion
The power plant of the cell—it releases energy from sugars.

Rough endoplasmic reticulum (ER)
Networks of ribosome-studded tubes, where proteins are manufactured.

Smooth endoplasmic reticulum
Tubes manufacturing fats and oils, and processing minerals.

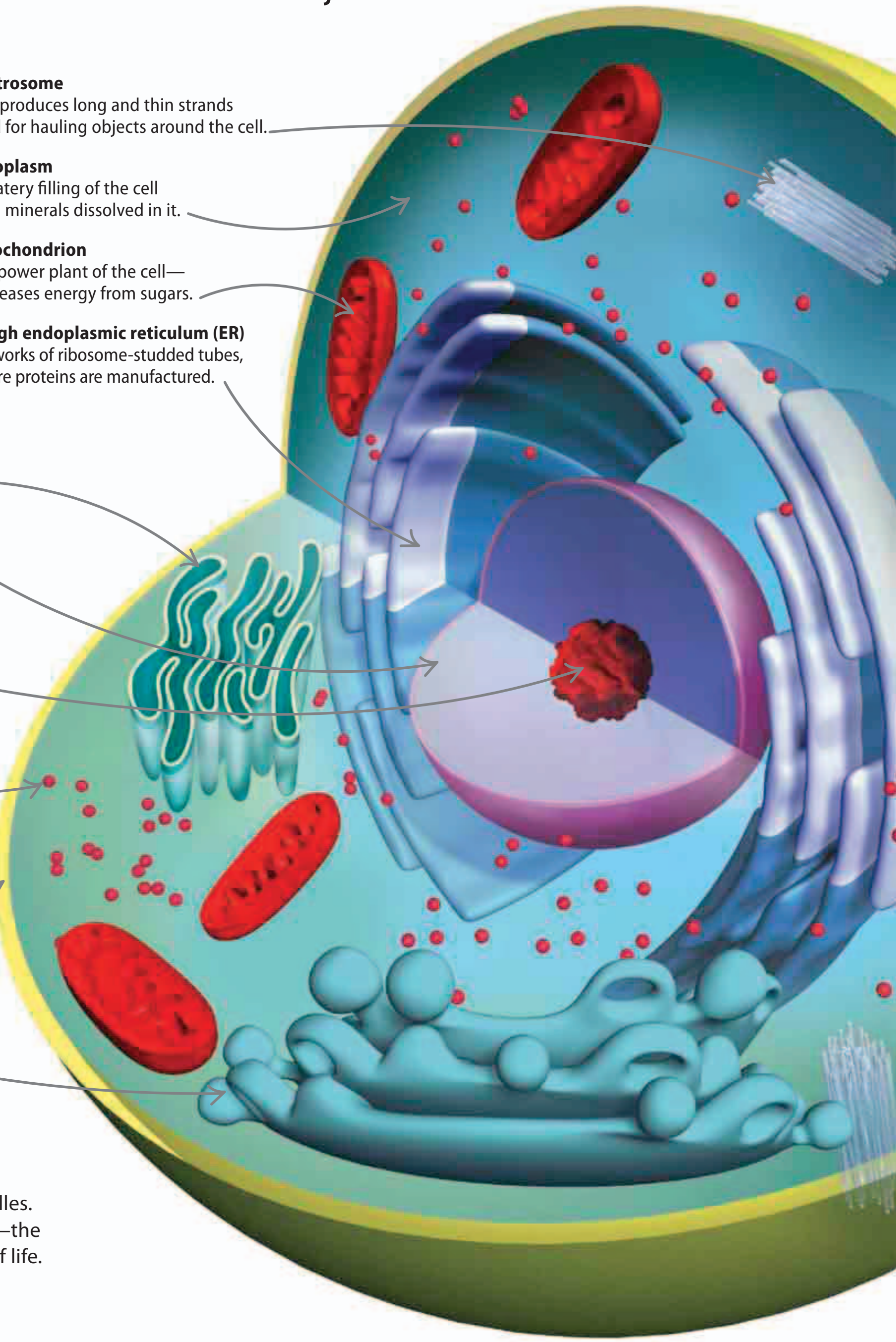
Nucleus
This contains the cell's genetic material, DNA—the instructions to build and maintain the cell.

Nucleolus
A dense region of the nucleus, which helps make ribosomes.

Ribosome
Genetic information in DNA is decoded here to make the proteins that build the cell.

Cell membrane
The selectively permeable outer layer through which certain substances pass in and out of the cell.

Golgi apparatus
Where newly made substances are packaged into membrane sacs, or vesicles, for transport around and out of the cell.



▷ **Animal cell construction**
The outer layer of most animal cells is a flexible membrane, which can take on any shape. The cell contains many types of tiny structures called organelles. Each one has a specific role in the cell's metabolism—the chemical processes necessary for the maintenance of life.